

INGEN DYNAMICS

From Portfolio Breadth to Platform Evidence

Eight-Week Applied AI Capstone Report

Competitive Strategy · Telemetry Analytics · Anomaly Detection · Reinforcement Learning · Decision Evaluation · Analyst Dashboard

CAPSTONE THESIS

InGen's most defensible opportunity is not any single robot. It is a shared physical-intelligence and evaluation layer that turns a broad portfolio into compounding evidence. The eight-week work supports a practical operating model: standardize telemetry and evaluation contracts across products, use specialized models for specific failure or decision regimes, preserve human governance, and require deployment evidence before capability claims.

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Executive Summary

SINGLE MOST IMPORTANT CONCLUSION

Portfolio breadth only becomes a strategic advantage when the products share a common evidence system. Across the eight-week work, the strongest recurring pattern was that reusable telemetry schemas, explicit evaluation contracts, calibrated uncertainty, human oversight, and reproducible deployment metrics create more durable value than choosing one “best” model. InGen should treat PIC 2.0 as the cross-portfolio control and evidence plane, while allowing each product to use the detector, policy, and decision logic best suited to its own operating regime.

The capstone began with a physical-AI landscape review and a 22-vendor competitive matrix spanning humanoid, service-robot, and security/patrol companies. The market is technically active but strategically fragmented: most competitors are concentrated on one embodiment, one workflow, or one vertical. That leaves InGen a credible opening around “one intelligence layer, many embodiments,” provided that shared architecture is visible in data, evaluation, safety, and deployment practice rather than only in positioning. Public robotics evidence also points to a similar reality: service robotics remains fragmented, adoption is use-case driven, and customers buy uptime, workflow fit, safety, and measurable return rather than foundation-model labels alone.[4][5]

Weeks 3 and 4 converted that strategic idea into an operational analytics stack for Aido Rover. A reproducible synthetic telemetry generator supports 50 units over 30 days at 1 Hz; the lightweight validation sample contains 5 units over 1 day, or 432,000 rows. The feature layer combines rolling statistics, cross-sensor residuals and ratios, context, missingness, and inter-unit z-scores. In a 10-split fault-injection benchmark, LOF was the strongest accuracy-maximizing detector with 0.985 ± 0.008 AUROC, 0.970 ± 0.033 recall, and 0.651 ± 0.055 F1. One-Class SVM was a close challenger. The capstone follow-up confirmed that the current LSTM Autoencoder should not be promoted for accelerated battery degradation: its AUROC was 0.566 ± 0.128 and it detected only 1 of 30 injected battery events at the original threshold. Battery-specific reconstruction channels carried more signal, but not enough to overturn the LOF recommendation.

Week 5 reinforced a second principle: model choice must be tied to operating constraints and behavior, not a single leaderboard metric. SAC produced much stronger and more stable BipedalWalker returns than PPO, but required roughly 12 times more wall-clock time. In multi-agent Simple Spread, shared-policy PPO improved average return and assignment distance relative to matched random behavior, yet increased collision frequency and never achieved the analyst-defined coordination threshold. These results are useful as design lessons for HTD-IRL and CRL-MRS, but they are simulation evidence, not proof of field-robot readiness.[16][17][18][20]

Week 6 showed the same distinction between repeatability and correctness in decision-model evaluation. The 50-scenario harness produced 150 seeded FLAN-T5-base judgments with a 100% parse rate. Overall nominal Krippendorff’s alpha was 0.725 and exact three-run agreement was 78%, but reliability varied sharply by domain, from 0.905 for Aido Rover to 0.366 for Senpai. The judge never passed a non-exact action, yet it passed only 59.7% of exact action matches. Therefore, exact action labels remain

authoritative and the LLM judge is best treated as a secondary review signal, not an autonomous scorer.[21][22]

Week 7 integrated the analytical outputs into a Streamlit dashboard for three personas. Engineering sees precision, recall, seeded stability and reliability first; Product sees learning trends, algorithm and runtime tradeoffs; Customer Success sees unit-level alert status and recommended action. The dashboard is deliberately evidence-aware: it distinguishes synthetic fleet status, benchmark results, and decision reliability instead of presenting all outputs as live operational truth.

Based on the evidence developed across the eight-week capstone, my recommendation is to organize InGen’s next phase around four cross-portfolio priorities: (1) a shared telemetry and provenance contract; (2) a common evaluation registry covering faults, policies, decision scenarios, thresholds, and seeds; (3) product-specific human-governance rules that can block or escalate uncertain actions; and (4) field metrics that measure operator workload, intervention, safety, uptime, and user outcomes. PIC 2.0 should make these capabilities reusable, while Fari, Senpai, Sentinel Prime AI, Aido Rover, and Aido Humanoid each retain product-specific policies and validation standards.

Capstone Scope and Evidence Map

The report integrates the eight-week internship as one analytical chain rather than eight disconnected deliverables. Weeks 1–2 define the strategic problem and portfolio opportunity. Weeks 3–4 establish a measurable Rover telemetry and anomaly stack. Week 5 tests learning-system tradeoffs in controlled RL benchmarks. Week 6 formalizes decision evaluation and seeded reliability. Week 7 turns the outputs into role-specific operational views. Week 8 adds the LSTM battery-degradation follow-up and converts the evidence into portfolio recommendations.

Table A. Eight-week capstone evidence chain.

Week	Artifact focus	Core question	Evidence produced
1	Landscape + PIC 2.0 map	What does modern physical AI require beyond a single model?	Foundation models, hierarchy, uncertainty, sim-to-real, deployment evidence
2	Competitive matrix + strategic gaps	Where can InGen differentiate?	Portfolio compounding, care/education, uncertainty-aware security
3	Telemetry generator + feature framework	What should be measured?	1 Hz synthetic fleet signals, context, rolling features, fleet-relative features
4	Anomaly benchmark	Which detectors are useful under different regimes?	10 seeded splits, five injected faults, LOF/OCSVM/IF/LSTM comparison
5	RL benchmark	What learning tradeoffs transfer to HTD-IRL / CRL-MRS?	Return, stability, runtime, sample efficiency, coordination behavior
6	Decision-eval harness	How reliable is model-based semantic scoring?	50 scenarios, 150 seeded judgments, alpha/kappa, criterion alignment
7	Streamlit dashboard	How should evidence be exposed to different users?	Engineering, Product and Customer Success control surfaces
8	Battery follow-up + capstone	Does the LSTM battery result change the recommendation?	No: LOF remains default; redesign battery sequence scoring before reuse

EVIDENCE BOUNDARY

Public-source claims are cited to external references. Quantitative fleet, anomaly, RL and decision-evaluation results are measured on synthetic telemetry, simulation environments, or synthetic scenarios unless explicitly stated otherwise. The report does not use production customer, patient, student or security data.

This evidence map also defines how conclusions should be read. The capstone supports comparative recommendations, evaluation design, and next-step priorities. It does not prove field deployment readiness. Where the evidence is indirect, the report labels the recommendation as an analyst inference rather than a validated product result.

1. Competitive Landscape and Strategic Gaps

The Week 2 competitor matrix profiled 22 vendors across three clusters: 10 humanoid companies, 6 service-robot companies, and 6 patrol/security vendors, with an as-of date of July 1, 2026. The analysis was designed to answer a strategic question rather than to rank every company: where can InGen create a position that is difficult to copy without matching both the portfolio and the shared intelligence layer?

The market signals from Week 1 point to increasing robotics adoption but uneven maturity. IFR reported 542,076 industrial robot installations in 2024 and more than 4.66 million industrial robots in operational stock; professional service-robot sales also increased, while the service-robot supplier base remained fragmented.[4][5] The research landscape is moving toward vision-language-action models, hierarchical control, and multi-rate architectures, yet the deployment bottleneck remains unchanged: a model that works in a paper or simulator still has to survive hardware variability, sensor noise, latency, safety constraints, and domain shift.[1][2][3][6]

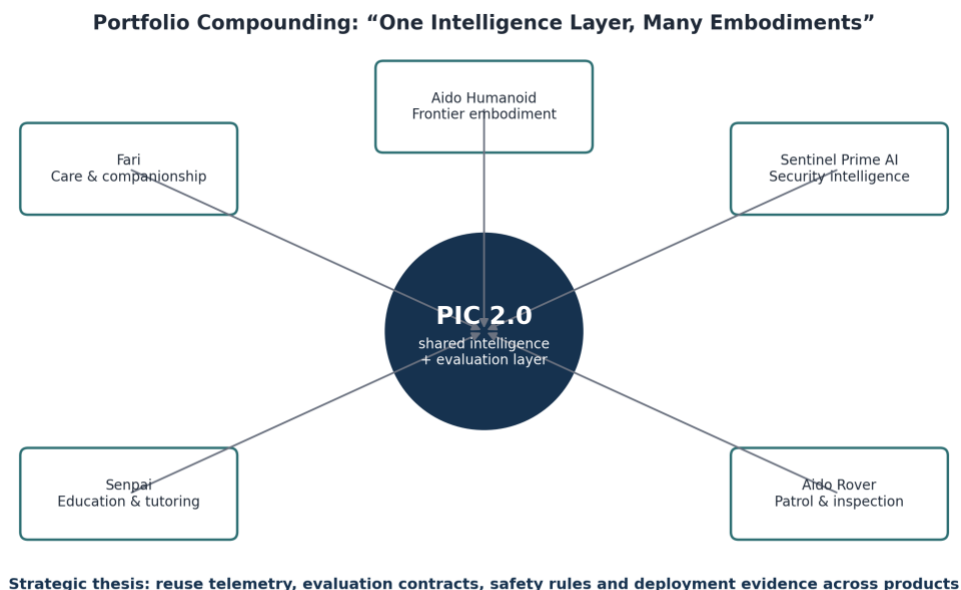


Figure 1. Portfolio compounding model for the InGen product set.

Source: synthesis of Weeks 1–2 and the PIC 2.0 conceptual map.

1.1 Strategic Gap 1: Fragmented products versus a reusable intelligence layer

The humanoid field is well funded but embodiment-led. Figure, 1X, Apptronik, Agility, Boston Dynamics, Sanctuary AI, and others are primarily scaling a humanoid or a focused labor platform. Figure's Helix is especially instructive because it separates slower semantic reasoning from high-frequency visuomotor control, underscoring that physical intelligence is likely to be layered rather than monolithic.[2] At the commercial level, funding rounds exceeding hundreds of millions or even a billion dollars reinforce how capital intensive the embodiment race has become.[10][11][12]

InGen's stronger position is not to claim a better general-purpose humanoid today. It is to make the portfolio itself an advantage: care, education, security, patrol, and future humanoid embodiments can share data contracts, evaluation infrastructure, governance, and deployment learning through PIC 2.0. The risk is equally clear. If each product develops its own telemetry, safety logic, metrics, and model interfaces, then breadth becomes duplicated engineering debt rather than compounding intelligence.

My Recommendation

I would measure PIC 2.0 reuse explicitly. For each new product release, the engineering review should state which schemas, evaluation harnesses, uncertainty modules, safety rules, and monitoring components are inherited unchanged, which are adapted, and which are product-specific. "Shared intelligence" should become a measurable reuse ratio, not only a platform narrative.

1.2 Strategic Gap 2: Human-centered care and education

Care and education are attractive because the social need is durable, but they are also domains where trust, governance, and longitudinal outcomes matter more than novelty. WHO projects the global population aged 60+ to reach 2.1 billion by 2050, while care-at-home robotics remains comparatively early.[7][5] Education faces a similar tension: AI can broaden access and personalization, but UNESCO and the World Economic Forum both emphasize the need for governance, human capability, and rapidly changing skills.[8][9]

Fari and Senpai therefore should not be positioned mainly around "AI companionship" or "AI tutoring." The defensible evidence is outcome-oriented: whether Fari can maintain useful engagement while escalating safely and respecting user autonomy, and whether Senpai can improve learning progress while remaining age-appropriate, auditable, and easy for educators or caregivers to override. The Week 6 decision harness already showed why this matters: judge reliability was lower for Fari and especially Senpai than for structured Rover decisions, so conversational ambiguity should be treated as an evaluation problem rather than hidden behind a single model score.

1.3 Strategic Gap 3: Security and patrol need uncertainty-aware operations

Security and patrol robotics compete on more than mobility. False alerts, operator attention, site-specific context, communications limits, and escalation policy are part of the product. IFR reported growth in security and inspection categories, but the operational differentiator is whether a system reduces workload without creating new noise.[5] The Sentinel Prime AI public framing around sensing,

classification, uncertainty gating, governance, and action therefore aligns with the right customer problem.[15]

The strongest portfolio combination is Sentinel Prime AI as the intelligence and escalation layer with Aido Rover as the mobile patrol and inspection layer. The Week 3–4 work demonstrates how that claim can be made measurable: Rover telemetry can expose energy, motor, GPS, connectivity, and software-health evidence; the anomaly stack can score it; and the dashboard can route the result to engineering or customer success. A security narrative grounded in alert precision, calibrated uncertainty, and operator workload is more defensible than a generic “autonomous patrol robot” narrative.

Table 1. Strategic-gap synthesis and evidence priorities.

Strategic gap	Market pattern	InGen opportunity	Evidence I would prioritize
Portfolio fragmentation	Humanoid and service vendors usually center on one embodiment or workflow	PIC 2.0 as shared data, evaluation, governance and learning layer	Track reuse across schemas, harnesses, safety rules and model interfaces
Care + education	Large social need; trust and outcome evidence remain hard	Fari + Senpai as human-centered adoption pathway	Measure engagement/learning outcomes, safe escalation and human override
Security + patrol	Alert quality and operator trust are as important as mobility	Sentinel Prime AI + Aido Rover	Optimize false-alert burden, uncertainty calibration and fleet follow-up

2. Telemetry Analytics Framework

The telemetry work was designed as the common measurement layer connecting product operation to later anomaly detection and dashboarding. The Week 3 generator supports 50 simulated Aido Rover units over 30 days at 1 Hz. For reproducible analysis, the main validation sample used 5 units over 1 day, producing 432,000 rows. Signals included battery state of charge and voltage, four drive-motor currents, motor temperature, IMU acceleration and gyroscope channels, Wi-Fi RSSI, GPS quality, task-success flags, mission mode, terrain, zone, weather, and light missingness.

The most important design choice was to keep raw telemetry and fault injection separate. Baseline telemetry includes benign operating variation such as charge cycles, thermal drift, terrain-linked motor load, RSSI multipath, GPS degradation, and measurement noise. Week 4 then injected controlled faults on top of this clean baseline. That separation makes the later benchmark auditable: the model is not rewarded for detecting anomalies that were accidentally baked into “normal” generation.

Telemetry Analytics Framework: From Signals to Decision Evidence

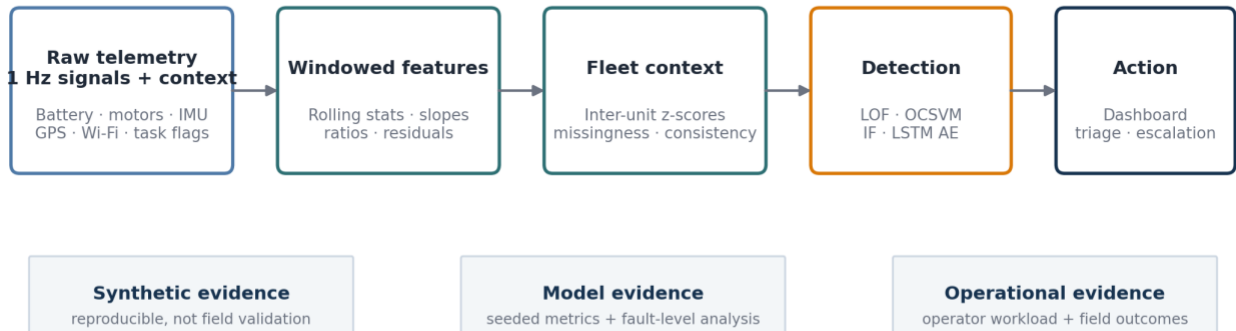


Figure 2. Telemetry analytics framework from raw signals to operational evidence.

Source: Weeks 3–4 synthesis.

2.1 Feature layers

The feature dictionary deliberately favored transparent transformations before complex modeling. Twenty-five engineered features covered short-window statistics, cross-sensor consistency, and fleet-relative context. Examples include five-minute SoC slope, battery-voltage-versus-SoC residual, motor side imbalance, motor temperature-to-current ratio, vibration RMS, speed-to-current efficiency, GPS/Wi-Fi consistency, task-success rate, missingness rate, and inter-unit z-scores.

Table 2. Feature-engineering layers used in the Rover analytics framework.

Feature layer	Representative features	Operational question
Energy	SoC mean/slope, discharge rate, voltage variability	Battery reserve, accelerated drain, charging-state transitions
Drive health	Mean current, rolling variability, side/front-rear imbalance	Motor stall, drag, asymmetric load, terrain effects
Motion / IMU	Accel/gyro norms, vibration RMS, gravity residual	Impact, instability, sensor drift, rough terrain
Localization + network	GPS HDOP/fix ratio, RSSI mean/std, GPS-Wi-Fi consistency	Jitter, degraded localization, communication risk
Fleet-relative context	Battery, motor-current and RSSI inter-unit z-scores	Peer-unit deviation at the same operational time
Data quality + outcome	Missingness rate, task-success rate	Software hang, telemetry degradation, mission reliability

2.2 What should become a cross-platform telemetry contract

The Rover schema is product-specific, but the deeper pattern is reusable. In my view, PIC 2.0 should define a small cross-platform telemetry envelope around every product event: timestamp, product/unit ID, software and model version, task or interaction ID, environment/context, confidence or uncertainty, action selected, human intervention, result, and data-quality flags. Product-specific sensor payloads can remain separate. The value comes from making evaluation, traceability, and fleet comparisons possible without forcing Fari, Senpai, Sentinel, Rover, and Humanoid into one physical sensor schema.

- Create a versioned “decision episode” identifier that links raw telemetry, model output, operator intervention, and outcome.
- Record model and policy versions alongside every action so regressions can be traced to a release rather than inferred from time alone.
- Keep context fields first-class. Terrain, zone, user state, bandwidth, battery reserve, or lesson stage often explain behavior better than a global score.
- Separate observed measurements from derived features and from model decisions. This preserves auditability and allows features to be recomputed later.
- Adopt explicit data-quality indicators so missingness or stale telemetry is not silently interpreted as a physical anomaly.

3. Anomaly Detection Recommendation

Week 4 compared Isolation Forest, Local Outlier Factor (LOF), One-Class SVM, and an LSTM Autoencoder across 10 seeded train/test splits. Five controlled fault types were injected only into held-out periods: motor stall, gradual sensor drift, accelerated battery degradation, GPS jitter/spoofing-like behavior, and intermittent software hang. Performance was measured with precision, recall, F1, and AUROC, with paired Wilcoxon signed-rank tests used to compare AUROC across matched seeds.

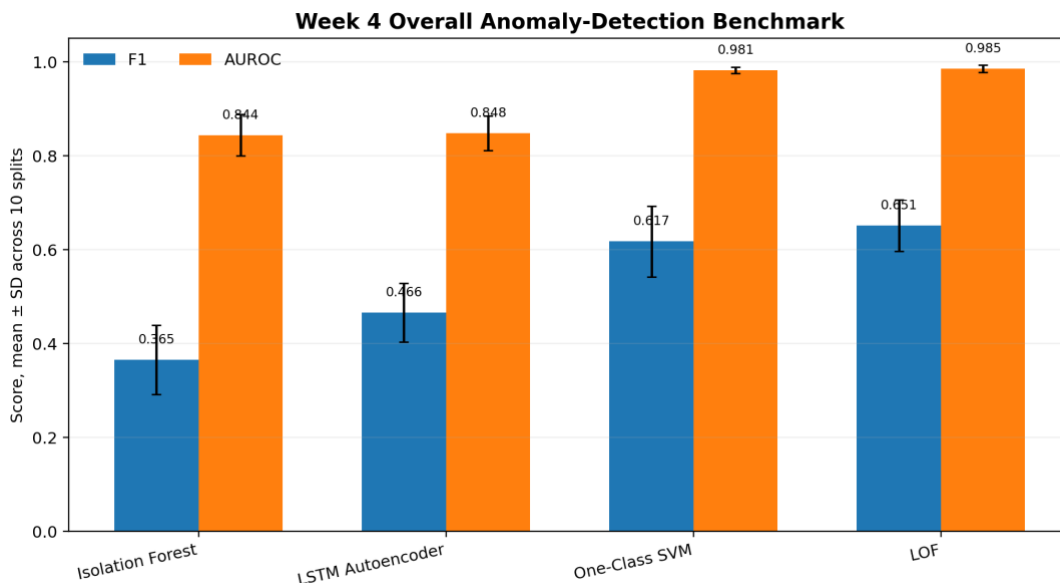


Figure 3. Overall anomaly-detection performance across 10 seeded splits.

Source: Week 4 benchmark; synthetic telemetry and injected faults.

Table 3. Overall Week 4 anomaly benchmark, mean $\pm$ SD.

Method	Precision	Recall	F1	AUROC
Isolation Forest	0.373 ± 0.129	0.394 ± 0.096	0.365 ± 0.074	0.844 ± 0.044
LSTM Autoencoder	0.382 ± 0.082	0.624 ± 0.069	0.466 ± 0.063	0.848 ± 0.037
One-Class SVM	0.458 ± 0.080	0.966 ± 0.022	0.617 ± 0.075	0.981 ± 0.007
LOF	0.492 ± 0.060	0.970 ± 0.033	0.651 ± 0.055	0.985 ± 0.008

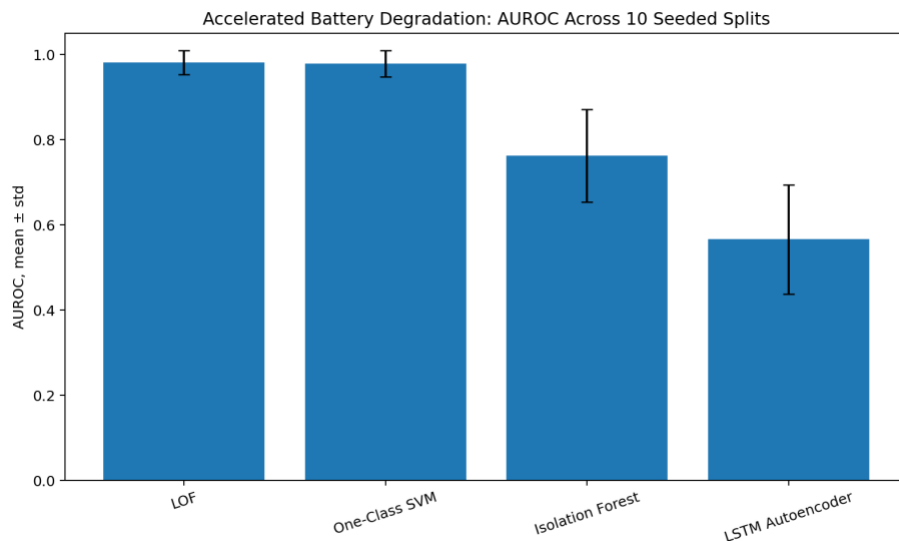
3.1 Recommended operating model

For accuracy-maximizing fleet analytics, LOF is the preferred detector. It had the strongest overall AUROC, recall, and F1, and it led four of the five individual fault categories. One-Class SVM was very close overall and effectively tied LOF on intermittent software hang, so it should remain a challenger model rather than be discarded. LOF's advantage over One-Class SVM was statistically detectable in the benchmark but operationally small; this is exactly the kind of gap that should be re-tested under larger and more heterogeneous telemetry before a production choice is locked.

For latency-constrained screening, Isolation Forest remains attractive because it is easier to score at the edge, but the benchmark does not support using it alone. Its overall recall was only 0.394. I would use it only as a first-pass screen paired with deterministic guards for battery-voltage consistency, stale telemetry, GPS quality, motor imbalance, and critical task failures. High-stakes escalation should combine a statistical detector with rule-based safety checks and explicit human review.

3.2 Follow-up: accelerated battery degradation

The specific analysis of the LSTM Autoencoder on accelerated battery degradation before the capstone was finalized. The follow-up reproduced the Week 4 behavior to floating-point tolerance and added event-level, threshold, training-duration, and feature-level reconstruction diagnostics.

**Figure 4. Accelerated-battery-degradation AUROC by detector.**

Source: Week 8 Follow-up; same 10 Week 4 splits.

The configured LSTM is not competitive on this fault. Its battery AUROC was 0.566 ± 0.128 , compared with 0.982 ± 0.028 for LOF and 0.979 ± 0.031 for One-Class SVM. Precision was 0.0016 ± 0.0049 , recall 0.0051 ± 0.0162 , and F1 0.0024 ± 0.0076 . Nine of ten splits had zero recall at the original threshold, and only 1 of 30 injected battery events was detected at least once.

The failure is not explained by one overly conservative threshold. Lowering the training-score threshold from the 97th to the 90th percentile increased mean battery recall to 0.149, but mean precision remained 0.013 and the clean-window flag rate rose to 18.5%. It is also not fixed by training for more epochs: 1, 5, and 10 epochs produced essentially unchanged AUROC. The strongest diagnostic was feature-level reconstruction: ``battery_voltage_soc_residual`` alone reached 0.783 ± 0.103 AUROC, while a four-channel battery-focused reconstruction score improved mean AUROC to 0.670 ± 0.093 . The model contains some battery signal, but equal-weight reconstruction error across 17 inputs dilutes it.

My Recommendation

Do not market the current LSTM Autoencoder as a specialized battery detector. Keep LOF as the default accuracy model. If battery sequence modeling is revisited, change the problem formulation: use hours- or charge-cycle-scale windows, explicit discharge-rate/capacity features, battery-weighted reconstruction scoring, more clean history, and an operator-calibrated alert threshold. A simpler dedicated battery-health baseline should be tested before adding more LSTM complexity.

4. Reinforcement-Learning Benchmark Findings and Transferable Lessons

Week 5 was intentionally framed as a learning benchmark, not a product-performance claim. PPO and SAC were evaluated on LunarLanderContinuous-v3 and BipedalWalker-v3 across five fixed seeds. A separate parameter-sharing PPO baseline was tested on three-agent Simple Spread. These environments are useful because they expose sample efficiency, instability, compute cost, and coordination behavior, but they do not reproduce Rover or Humanoid dynamics. This evidence boundary is important: sim-to-real remains a known robotics bottleneck, and toy-environment success should never be treated as deployment evidence.[6]

Week 5 RL Benchmark: Learning Quality vs. Compute Cost

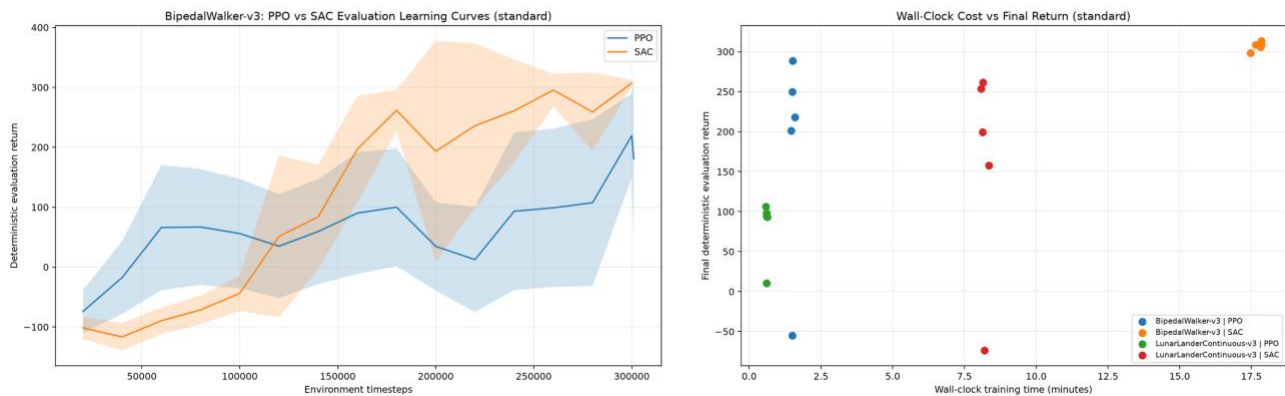


Figure 5. Week 5 RL benchmark: BipedalWalker learning dynamics and wall-clock tradeoff.

Source: Week 5 five-seed benchmark; CPU timing on Apple Silicon.

Table 4. Single-agent RL results, mean $\pm$ SD across five seeds.

Environment	Algorithm	Final return	Threshold	Wall-clock	Interpretation
BipedalWalker	PPO	180.57 $\pm$ 135.76	0/5	91.0 s	Fast but unstable; one late regression
BipedalWalker	SAC	306.96 $\pm$ 5.50	4/5	1,063.3 s	Strong and stable; $\sim 11.7\times$ slower
LunarLander	PPO	80.46 $\pm$ 39.55	0/5	36.2 s	Cheap baseline; did not reach threshold
LunarLander	SAC	159.75 $\pm$ 137.17	3/5	490.9 s	Higher upside; seed-sensitive

4.1 Lesson 1: Match the learning regime to the hierarchy level

PPO and SAC expose a useful design tradeoff. PPO is on-policy and comparatively cheap in these runs; SAC reuses experience off-policy and achieved much stronger BipedalWalker performance, but at a large compute cost.[16][17] The transferable lesson for HTD-IRL is not “use SAC.” It is to define separate optimization and evaluation contracts for different hierarchy levels. High-level planning can tolerate slower updates and should be measured on subgoal quality and plan consistency. Low-level control needs fast, robust, safety-constrained behavior. The interface between levels must be versioned so a lower-level controller update does not silently invalidate a higher-level policy.[18]

4.2 Lesson 2: Behavior and stability matter more than aggregate reward

Two benchmark outcomes show why a single scalar reward is insufficient. BipedalWalker PPO seed 47 reached a strong intermediate checkpoint and then ended with a negative final return, demonstrating checkpoint regression. In Simple Spread, trained PPO improved return and assignment distance but increased collisions and did not achieve the coordination threshold. An optimizer can therefore improve the metric it sees while degrading a behavior the product actually cares about.

For physical AI, I would make policy promotion conditional on a bundle of metrics: task completion, collision/safety violations, energy and time cost, intervention frequency, subgoal churn, uncertainty, worst-scenario behavior, and regression relative to the incumbent. CRL-MRS should add retention and rollback tests so a fleet update cannot improve the mean while forgetting previously reliable behavior. This is consistent with continual-RL research, where non-stationarity and retention are part of the problem rather than noise around a final score.[19]

4.3 Lesson 3: Shared policies are a baseline, not a coordination solution

In Simple Spread, shared-policy PPO improved mean agent return from -23.73 to -21.16 and reduced final assignment distance from 0.919 to 0.682. The improvement in assignment distance occurred in all five seeds, but neither PPO nor the random baseline achieved the analyst-defined coordination threshold. PPO also increased average collision pairs per cycle from 0.046 to 0.074. With only five paired seeds, the Wilcoxon tests did not support statistical claims.

For CRL-MRS, the next benchmark should include at least one true multi-agent baseline with explicit centralized-training/decentralized-execution structure, role or capability context, and clearer credit assignment. Parameter sharing is useful when robots are interchangeable; it becomes fragile when battery health, sensor quality, payload, localization confidence, or assigned role differ. The benchmark supports the need for explicit coordination structure, not a claim that one MARL architecture is already selected.[20]

5. Decision-Model Evaluation Methodology and Reliability

Week 6 created a 50-scenario decision-evaluation harness covering Aido Rover, Sentinel Prime AI, Fari, and Senpai. Each YAML scenario defines structured inputs, allowed actions, an authoritative target action, a target rationale, risk metadata, and a publication-safe synthetic context. A rule-based controller provides a transparent baseline. A small open Hugging Face model, `google/flan-t5-base`, provides a secondary PASS/PARTIAL/FAIL rubric across judge seeds 7, 17, and 27.

The architecture deliberately separates correctness from semantic review. Exact match against `target\_action` is authoritative. The LLM judge can help expose ambiguity or rationale mismatch, but it cannot redefine the answer. This avoids a circular system where the same model both creates and grades its own behavior.

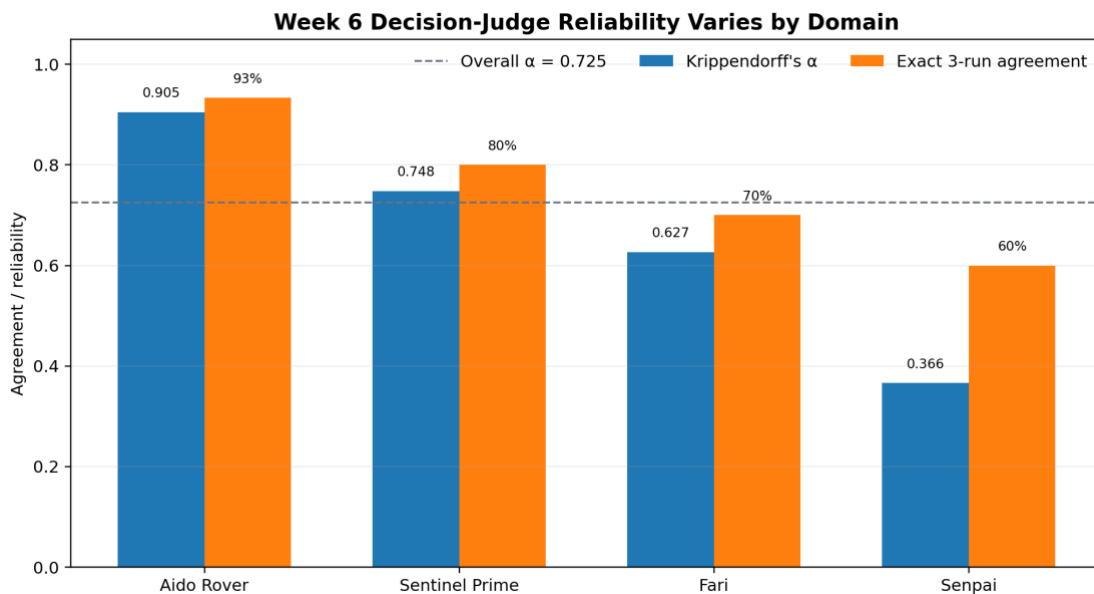


Figure 6. Decision-judge reliability by product domain.

Source: Week 6, 50 scenarios $\times$ 3 seeded judge runs.

Table 5. Week 6 reliability summary.

Scope	Scenarios	Judgments	3-run agreement	Krippendorff α	Interpretation
Overall	50	150	78.0%	0.725	Tentative; use with review
Aido Rover	15	45	93.3%	0.905	Strong for this test bank

Scope	Scenarios	Judgments	3-run agreement	Krippendorff α	Interpretation
Sentinel Prime AI	15	45	80.0%	0.748	Tentative
Fari	10	30	70.0%	0.627	Limited
Senpai	10	30	60.0%	0.366	Poor for automated use

5.1 Reliability is necessary but not sufficient

Overall nominal Krippendorff's alpha was 0.725 and pairwise Cohen kappa ranged from 0.621 to 0.817. Those values show that the seeded judge has meaningful repeatability, especially for structured Rover scenarios.[21][22] However, criterion alignment exposes an equally important limitation: among 129 exact action matches across the three seeded runs, only 77 received PASS. The resulting exact-match PASS recall was 59.7%. At the same time, the judge never passed any of the 21 non-exact actions. The judge is conservative, but conservatism is not the same as correctness.

The domain pattern is strategically useful. Rover decisions were the most repeatable, while Senpai was the least stable. That does not mean the Rover product is "better" than Senpai; it means structured route/energy decisions are easier for the current judge than conversational tutoring and safeguarding decisions. The right response is not to tune the prompt until every domain receives a high alpha. It is to route ambiguous, human-centered decisions toward deterministic safeguards, domain-owner review, or stronger human-labeled evaluation.

5.2 What PIC 2.0 should reuse from the harness

- Versioned scenario schema: inputs, allowed actions, target behavior, rationale, risk level and source/provenance.
- Authoritative scoring contract separated from optional model-based semantic review.
- Multi-seed evaluation by default for stochastic judges or policies.
- Domain-level reliability rather than a single portfolio-wide average.
- A queue of disagreement and high-risk cases for human review.
- Release metadata linking model/prompt version, seed, threshold and result artifact.

My Recommendation

The decision harness is one of the most transferable Week 6 artifacts. I would evolve it into a PIC 2.0 evaluation registry where every platform can register scenarios, regression tests, thresholds, safety rules and expected actions. The key is not a universal judge; it is a universal contract for proving what changed and whether the change is acceptable.

6. Dashboard Summary: Turning Benchmarks into Role-Specific Decisions

Week 7 synthesized Weeks 3–6 into a Streamlit dashboard with three workspaces: Fleet Health, Policy Benchmark, and Decision Evaluation. The design objective was not to create one dense technical page. It was to make the same evidence usable by three people who ask different questions.

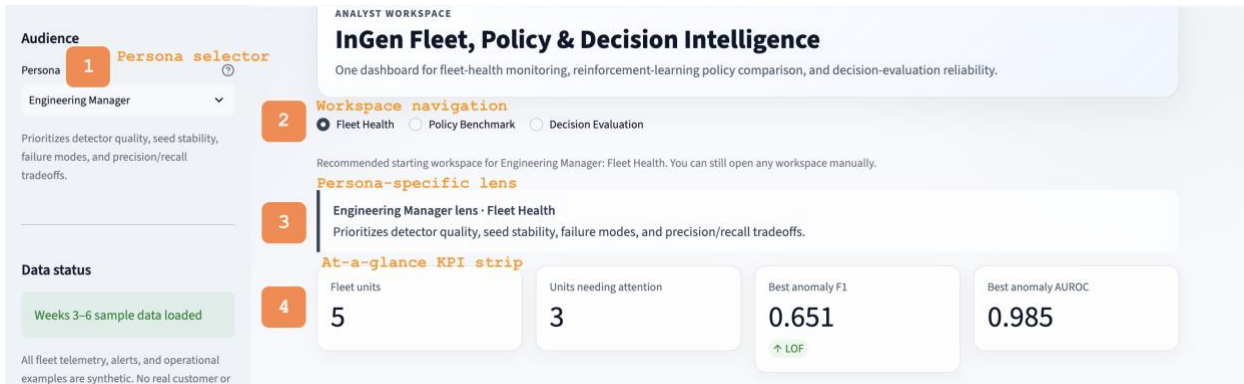


Figure 7. Persona-aware dashboard navigation and executive KPI surface.

Source: final Week 7 Streamlit dashboard.

Table 6. Persona-to-control mapping in the Week 7 dashboard.

Persona	Default workspace	Primary decision	Control surface
Engineering Manager	Fleet Health	Which detector is accurate and stable? Which faults or scenarios regress?	Precision, recall, F1, AUROC, split stability, reliability drill-down
Product Manager	Policy Benchmark	Which method performs best, at what compute cost, and with what readiness caveat?	Learning curves, final reward, wall-clock, threshold success, cross-platform scorecards
Customer Success	Fleet Health	Which unit needs attention and what should happen next?	Status, battery, active alert, severity, recent trends, recommended action

The dashboard is intentionally persona-aware but not role-restrictive. Changing the persona changes the recommended starting workspace and content ordering; users can still inspect other workspaces. This matters because product, engineering, and customer-success decisions overlap. A customer escalation may require a model-quality drill-down, and a product decision may require operational alert evidence.

The more important design principle is evidence labeling. Fleet-unit status is synthetic. RL results are simulation benchmarks. Decision reliability is a judge-quality measure, not product correctness. Keeping those boundaries visible reduces the risk that a polished dashboard turns experimental results into accidental production claims.

7. Forward-Looking Suggestions for the InGen Portfolio

The recommendations below are my proposed priorities based on the competitive landscape, the synthetic telemetry and anomaly work, the RL benchmark, the decision-evaluation reliability analysis, and the dashboard integration.

7.1 PIC 2.0: Make the shared layer an evidence and governance plane

PIC 2.0 should be the place where cross-product contracts live: telemetry provenance, model and policy versions, evaluation scenarios, uncertainty fields, safety constraints, release gates, and rollback metadata. I would resist the temptation to make PIC 2.0 synonymous with one foundation model. The Week 1 conceptual map is stronger when interpreted as a layered physical-intelligence control stack: optimization, uncertainty, safety, decision control, hierarchy, and continual multi-robot learning. The cross-product advantage comes from standardized interfaces and evidence, while product policies remain specialized.

- Create a cross-product schema registry and evaluation registry.
- Require release artifacts that connect model version → benchmark → thresholds → known failure modes.
- Use uncertainty and human-override fields consistently across Sentinel, Rover, Fari and Senpai.
- Track reuse as an engineering KPI: shared modules should reduce validation time for the next product rather than only increase architectural complexity.

7.2 Aido Rover: Treat fleet health as a first-class product capability

Rover is the platform where the eight-week analytical stack is most concrete. It already has a synthetic telemetry contract, engineered fleet-health features, injected-fault benchmarks, a clear LOF recommendation, and a customer-success dashboard. I would turn that into a persistent fleet-health service rather than a one-time model. The service should score health, explain which channels drove the alert, preserve route and battery context, and support operator acknowledgement and feedback.

- Use LOF as the default accuracy benchmark and keep One-Class SVM as a challenger.
- Add deterministic guards for low reserve, voltage-versus-SoC inconsistency, stale telemetry and localization failure.
- Build battery health on longer charge-cycle history; do not reuse the current LSTM result as evidence of readiness.
- Connect health state to route arbitration so an energy or drivetrain warning changes the allowable mission set rather than only producing a dashboard alert.

7.3 Sentinel Prime AI: Optimize operator trust, not just detector sensitivity

Sentinel should be measured on the downstream operator system, not only on raw model accuracy. The right metric bundle includes precision, calibrated confidence, false-alert burden per hour, time-to-review, escalation rate, evidence completeness, and the fraction of alerts overturned by humans. The Week 6 Sentinel scenarios and the Week 7 dashboard already model bandwidth-aware escalation and decision review; those should become regression tests around product changes.

- Create alert budgets by site and severity so threshold changes are evaluated against operator capacity.
- Log the uncertainty and evidence bundle that caused every escalation.

- Use Rover as a mobile follow-up layer only after fixed-sensor confidence and patrol value are established.
- Treat human override and auditability as product features that can be measured and demonstrated.

7.4 Fari: Build trust around autonomy boundaries and caregiver escalation

For Fari, the most important technical challenge is not maximizing conversation length. It is knowing when to continue, when to ask, and when to escalate. The Week 6 judge was less reliable on Fari than on Rover, which is a useful warning that care decisions contain semantic and social ambiguity. I would use deterministic safety rules and a clearly defined escalation ladder around health, medication, distress, consent, and caregiver contact, with the generative model operating inside those boundaries.

- Measure healthy engagement and unnecessary escalation separately.
- Create human-reviewed scenario sets for refusal, uncertainty, medication, privacy and emergency handoff.
- Store user preferences and consent separately from inferred model state.
- Pilot with outcome metrics such as engagement, caregiver burden, successful handoff and intervention rate rather than only conversation quality.

7.5 Senpai: Make learning outcomes and safeguarding the product-level scorecard

Senpai produced the weakest LLM-judge reliability in the Week 6 harness, with alpha 0.366. That result belongs to the evaluation judge rather than the Senpai product, but it highlights the ambiguity of tutoring and safeguarding decisions. I would avoid using one generative judge as the main evaluation mechanism. Curriculum progress, misconception correction, age-appropriate behavior, intervention policy, and teacher or guardian override should be measured through explicit rubrics and human-reviewed cases.

- Use curriculum-linked learning objectives and pre/post evidence where feasible.
- Gate sensitive or safeguarding branches with deterministic policies and human escalation.
- Separate pedagogical adaptation from emotional or safety decisions so each has its own evaluation contract.
- Use the scenario harness to build a regression suite around edge cases, not to replace educator judgment.

7.6 Aido Humanoid: Follow an evidence ladder from skills to embodied deployment

Aido Humanoid is the most ambitious embodiment and the least directly validated by the eight-week experiments. The correct recommendation is therefore methodological. Helix and the broader physical-AI literature suggest multi-rate control: slower semantic/task reasoning and faster reactive control loops.[2] The RL benchmark shows why stability, collisions, checkpoint regression and compute cost must be part of evaluation. I would build a staged evidence ladder before making general-purpose autonomy claims.

- Start with a small library of bounded skills with clear initiation, termination and failure conditions.

- Use simulation for iteration, but require sim-to-real tests that include sensor noise, latency, calibration and hardware variation.[6]
- Measure intervention rate and safety constraints alongside task success.
- Treat cross-embodiment transfer as a hypothesis to test through standardized tasks rather than an assumed benefit of a shared foundation model.

7.7 Cross-portfolio priority sequence

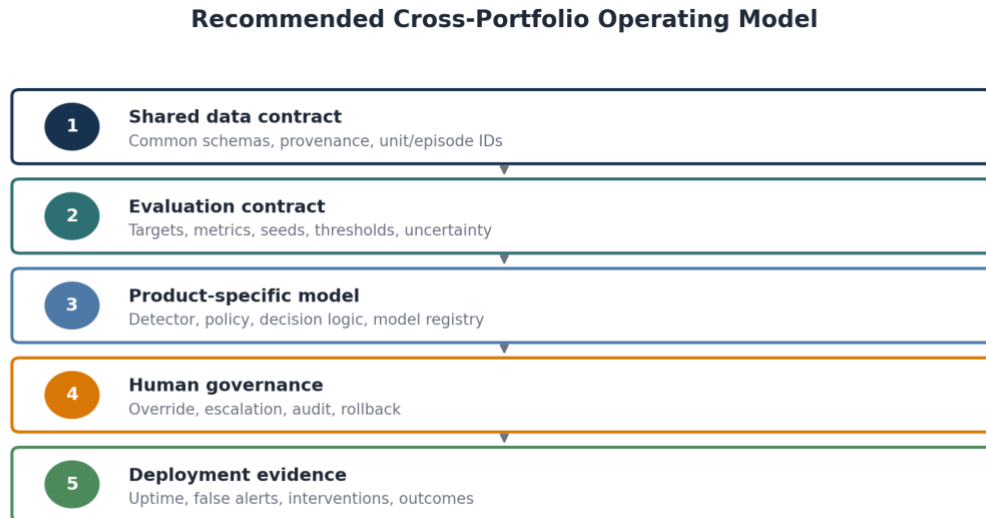


Figure 8. Recommended cross-portfolio operating model.

Source: analyst synthesis of the eight-week capstone.

The priority order is intentional. Data and evaluation contracts come before a larger model portfolio; product-specific model choice comes before centralized “one-model-fits-all” claims; governance comes before autonomous escalation; and deployment evidence comes before strategic scaling. This sequence creates a common spine without erasing domain differences.

Table 7. Platform-specific recommendations from the analyst perspective.

Platform	Next priority	Why it matters	Suggested evidence
PIC 2.0	Shared telemetry + evaluation registry	Cross-product reuse and release traceability	% reused modules; regression pass rate; rollback readiness
Aido Rover	Fleet-health service + energy-aware action constraints	Operational reliability and patrol uptime	Alert precision/recall; intervention; uptime; energy reserve
Sentinel Prime AI	Uncertainty-aware alert operations	Lower operator burden with auditable escalation	False alerts/hour; ECE; review time; override rate
Fari	Safe autonomy boundary + caregiver escalation	Trustworthy care support	Engagement; handoff success; unnecessary escalation; consent violations
Senpai	Learning outcomes + safeguarding regression	Useful tutoring with auditable safety	Learning gain; misconception correction; intervention;

Platform	Next priority	Why it matters	Suggested evidence
	suite		safeguard pass rate
Aido Humanoid	Skill-based evidence ladder + sim-to-real discipline	Credible progress toward general embodied autonomy	Task success; interventions; collisions; transfer gap; runtime

7.8 Recommended 12-Month Validation Sequence

If this eight-week body of work were extended into a 12-month validation program, I would not expand every research track in parallel. I would sequence the work so that shared infrastructure is established first, two operationally concrete products generate evidence next, and human-centered or frontier embodiments deepen after the evaluation spine is stable.

Table 8. Suggested 12-month sequencing from the analyst perspective.

Horizon	Program focus	Primary platforms	What I would build	Exit criterion
0–3 months	Standardize the spine	PIC 2.0 + all products	Versioned telemetry/decision episode schema; evaluation registry; release metadata; common dashboard definitions	A new experiment can be reproduced and compared without bespoke data plumbing
3–6 months	Operationalize Rover + Sentinel	Aido Rover, Sentinel Prime AI	Longer fleet telemetry; field-shaped fault taxonomy; alert budget; operator review workflow; battery-history redesign	Alert quality, operator workload and fleet-health evidence are measurable end to end
6–9 months	Human-reviewed decision evidence	Fari, Senpai	Domain-owner scenario labels; consent/safeguarding escalation tests; outcome rubrics; disagreement review	Generative behavior is bounded by explicit human-centered contracts
9–12 months	Embodied hierarchy + transfer	Aido Humanoid + PIC 2.0	Skill library, high/low-level policy interface, sim-to-real stress tests, regression and rollback	Humanoid claims are tied to bounded skills and measured transfer gaps

WHY THIS ORDER

The fastest way to lose the portfolio advantage would be to start six separate model programs. The fastest way to compound it is to make every experiment use the same evidence spine, then let product-specific models compete inside that shared measurement and governance framework.

8. Key Limitations and What the Evidence Does Not Prove

The capstone is intentionally evidence-bounded. The strongest parts are reproducibility, controlled comparisons, and explicit contracts. The weakest boundary is external validity: most quantitative work uses synthetic telemetry, benchmark environments, or synthetic decision scenarios rather than production data.

Table 9. Evidence boundaries and next validation steps.

Limitation	Current boundary	Next validation step
Synthetic fleet data	Rover telemetry and fault events are controlled simulations	Re-run on longer field logs and real incident labels before deployment claims
Small time horizon	Main Rover benchmark uses 5 units $\times$ 1 day	Expand across sites, seasons, routes, software versions and charge cycles
Unsupervised anomaly threshold	Fixed training-score quantiles are not operator-calibrated	Tune against false-alert workload and safety cost
RL simulation	LunarLander, BipedalWalker and Simple Spread are teaching benchmarks	Add product-shaped simulators, safety constraints and sim-to-real validation
Decision labels	Targets are analyst-defined synthetic contracts	Add domain-owner/human-reviewed labels and adjudication
Small open judge	FLAN-T5-base is conservative and domain-sensitive	Compare stronger open judges or human review, but preserve authoritative contracts
Dashboard snapshot	Local CSV dashboard is not a live operations system	Add refresh, authentication, alert acknowledgement and audit trail

The practical consequence is that the capstone should be used to guide the next experiment and the next product metric, not to claim that the portfolio has already solved anomaly detection, multi-robot coordination, or autonomous human-centered decision-making. This restraint is a strength. A physical-AI platform becomes credible when it has a repeatable path from concept $\rightarrow$ synthetic benchmark $\rightarrow$ simulation $\rightarrow$ pilot $\rightarrow$ monitored deployment $\rightarrow$ post-deployment recalibration, with clear gates between each stage.[6]

9. Closing View

Across the eight weeks, the most consistent lesson was that physical AI is an evidence-integration problem as much as a model problem. Foundation models can improve semantic generalization; RL can improve policies; anomaly models can surface unusual behavior; LLM judges can add semantic review. None of those components is sufficient by itself. The product has to know what data produced the decision, how confident the system was, what safety rules applied, whether a human intervened, and whether the outcome improved.

That is why the capstone recommendation is centered on PIC 2.0 as a shared evidence and governance plane rather than a universal algorithm. Aido Rover should deepen fleet-health evidence. Sentinel should optimize alert quality and operator trust. Fari and Senpai should formalize human-centered boundaries and outcome metrics. Aido Humanoid should progress through a staged skill and sim-to-real evidence ladder. The shared layer should make every one of those programs easier to evaluate, compare, audit, and improve.

Final Recommendation

Build the next phase around reusable contracts and measurable outcomes: one cross-portfolio data/evaluation spine, product-specific models and safeguards, and a disciplined evidence ladder from synthetic benchmarks to field outcomes. That approach converts InGen's breadth from a strategic risk into a compounding platform advantage.

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Internal Capstone Evidence Base

- Week 1: Physical AI Landscape Brief and PIC 2.0 Conceptual Map.
- Week 2: 22-vendor Competitive Matrix and Strategic Gap Memo.
- Week 3: Aido Rover Telemetry Generator, EDA, and Feature Dictionary.
- Week 4: Anomaly Benchmark, Method Recommendation Memo, and midpoint synthesis.
- Week 5: PPO/SAC and multi-agent RL benchmark results and analyst memo.
- Week 6: 50-scenario Decision Evaluation Harness and reliability methodology.
- Week 7: Streamlit dashboard, persona design document, sample data, and narrated walkthrough.
- Week 8: LSTM accelerated-battery-degradation follow-up.

Evidence note: all quantitative fleet-health results in this report are synthetic or benchmark-derived unless explicitly described as public external market evidence. No production customer, patient, student, or security data is used.